

Machinery Dynamics—Fundamentals (Ch. 10)

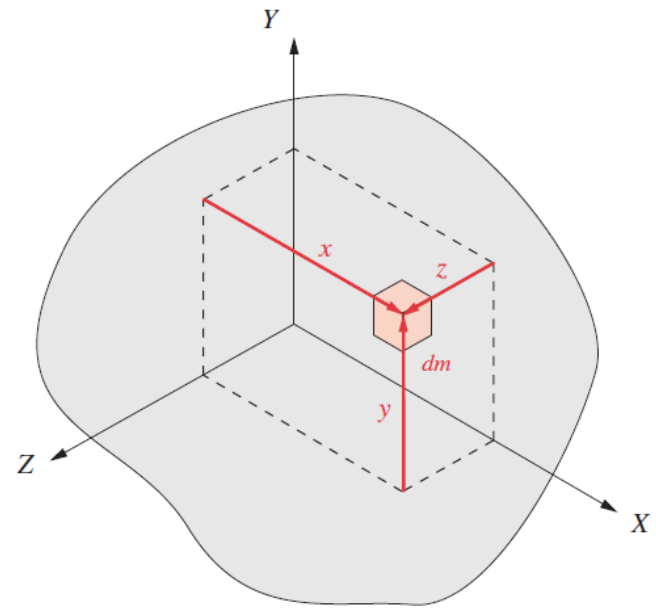
- **Newton's Laws of *particle* motion**
 - Inertia property of motion (**Law 1**)
 - $\mathbf{F} = m\mathbf{a} = d\mathbf{M} / dt$ $\mathbf{M} = m\mathbf{v}$, (**Law 2**)
 - $\mathbf{F}_1 = -\mathbf{F}_2$ (**Law 3**)

• Inertial parameters

- Mass

$$m = \int dm$$

- Centre of Mass/Gravity (COM/COG)

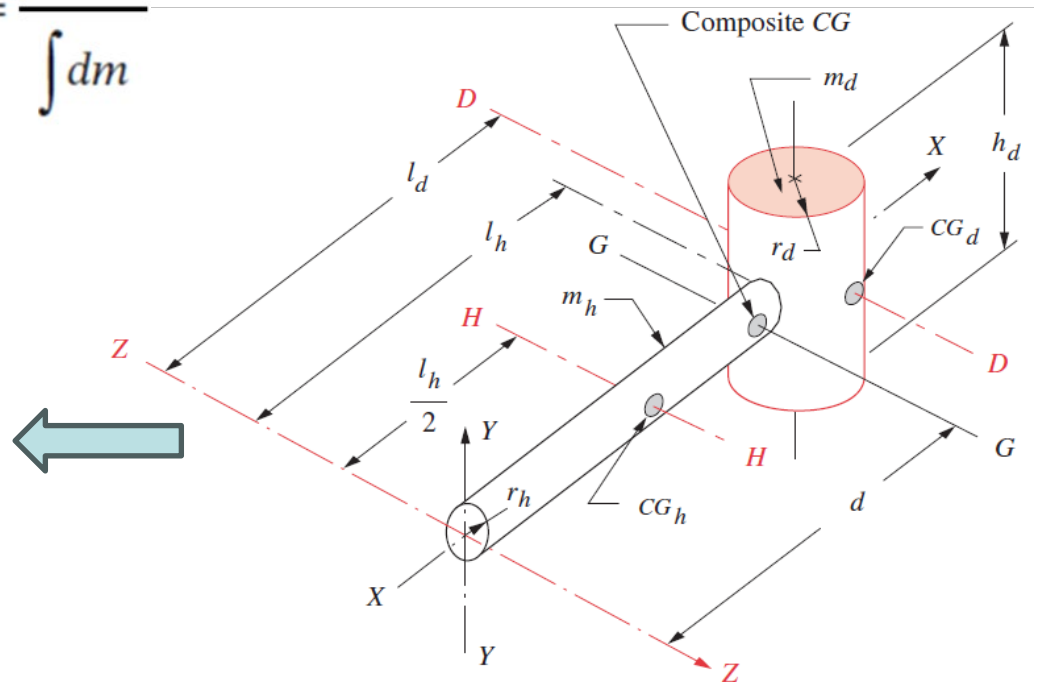


$$X = \frac{\int x dm}{\int dm}$$

$$Y = \frac{\int y dm}{\int dm}$$

$$Z = \frac{\int z dm}{\int dm}$$

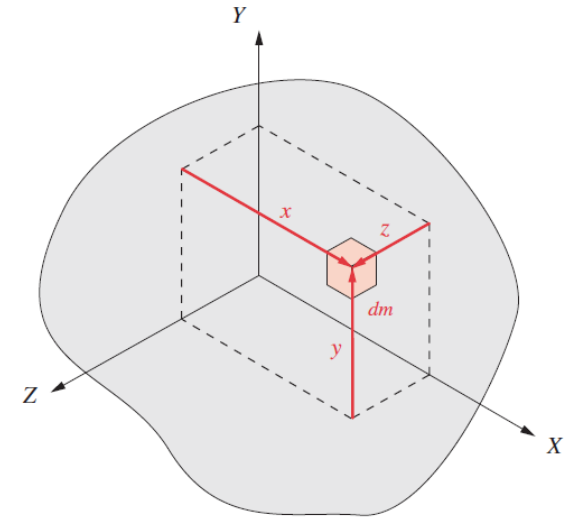
$$d = \frac{\int x dm}{\int dm} = \frac{m_h \frac{l_h}{2} + m_d l_d}{m_h + m_d}$$



– Moment of inertia

$$I_x = \int (y^2 + z^2) dm \quad I_y = \int (x^2 + z^2) dm$$

$$I_z = \int (x^2 + y^2) dm$$



– Parallel axis theorem

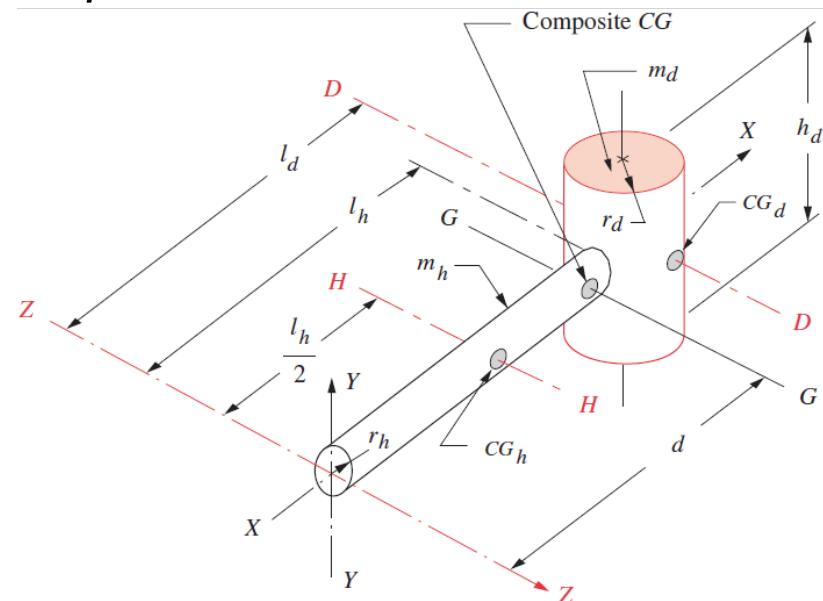
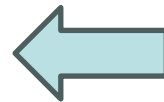
It relates the moment of inertia (MOI) of a body about an *axis passing through its COM* to the MOI of a *parallel axis*

$$I_{ZZ} = I_{GG} + md^2$$

$$I_{HH} = \frac{m_h (3r_h^2 + l_h^2)}{12}$$

$$I_{DD} = \frac{m_d (3r_d^2 + h_d^2)}{12}$$

$$I_{ZZ} = \left[I_{HH} + m_h \left(\frac{l_h}{2} \right)^2 \right] + \left[I_{DD} + m_d l_d^2 \right]$$



– Radius of gyration

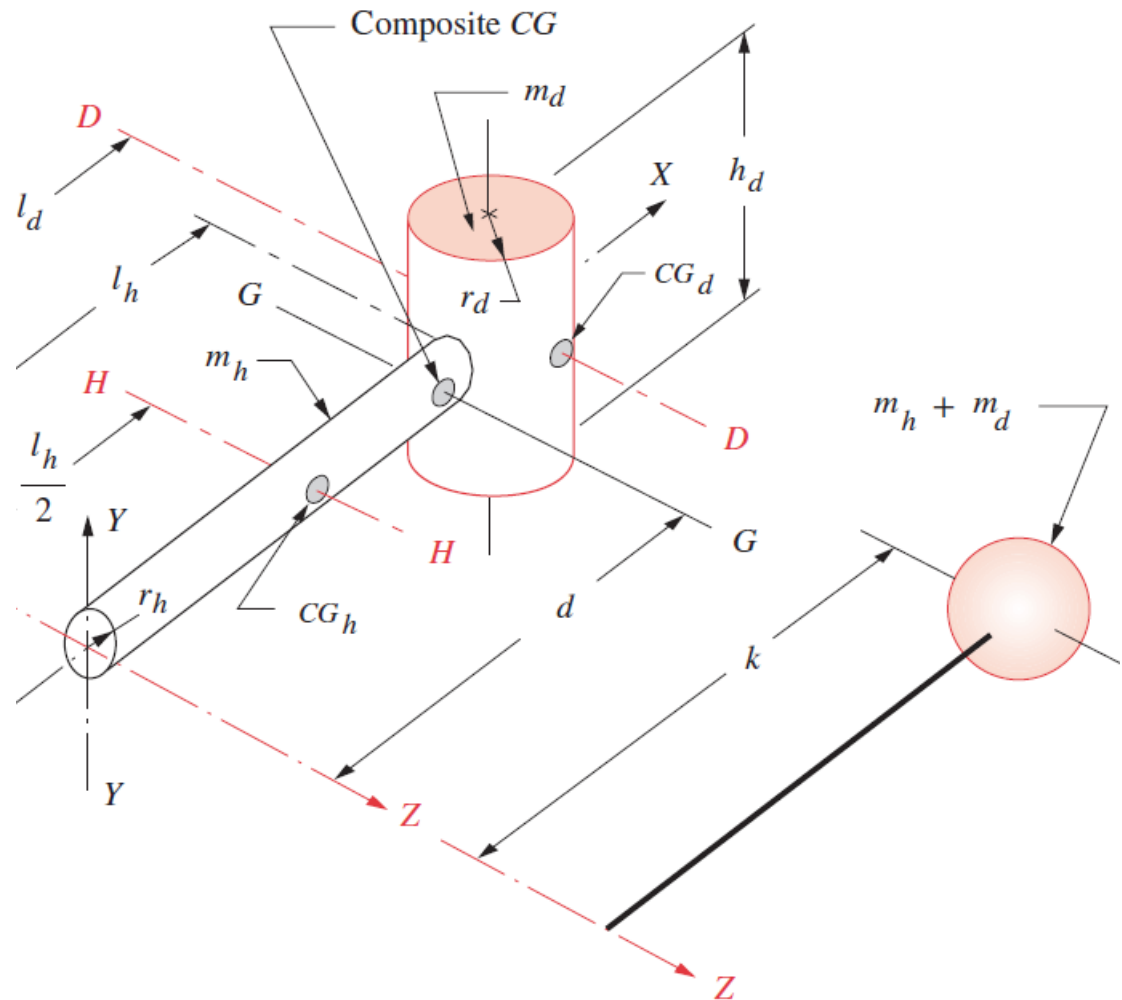
The radius at which the entire mass of the body is concentrated such that the *resulting moment of inertia* (MOI) is the same as that of the original body

$$I_{ZZ} = mk^2$$

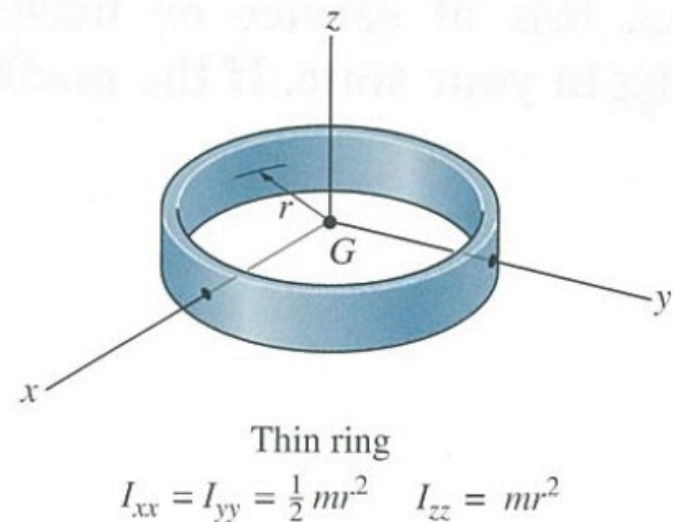
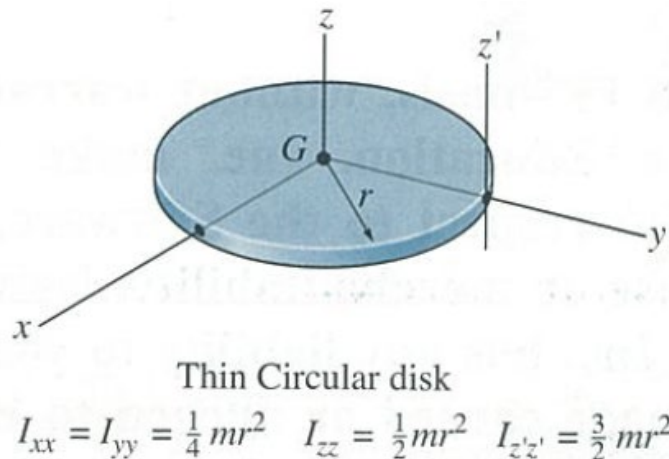
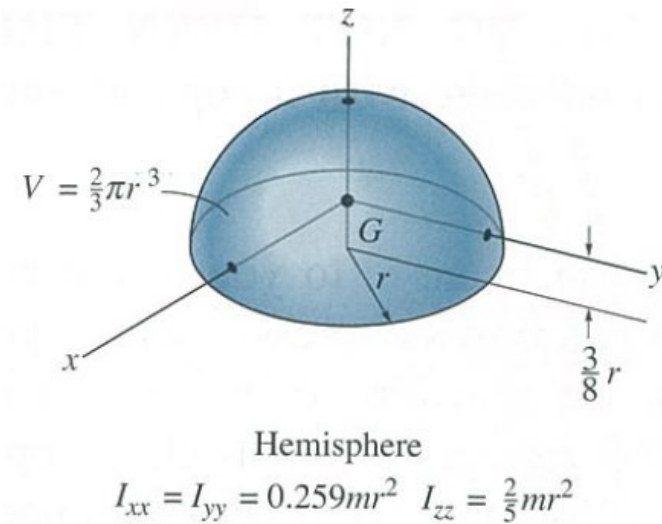
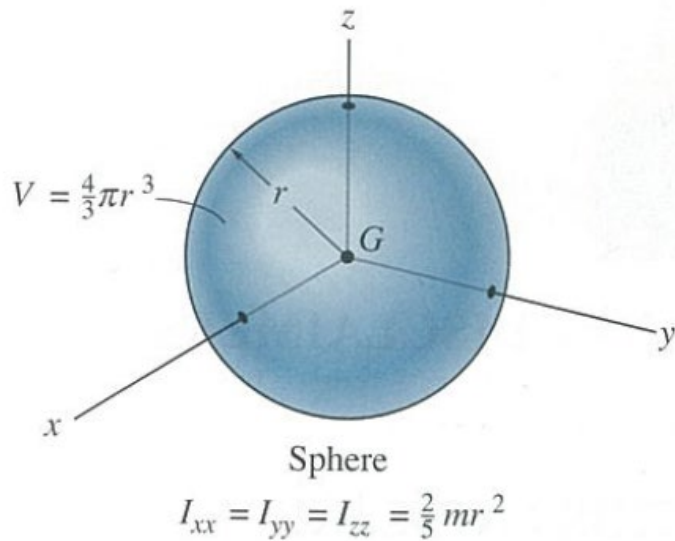
$$k = \sqrt{\frac{I_{ZZ}}{m}}$$

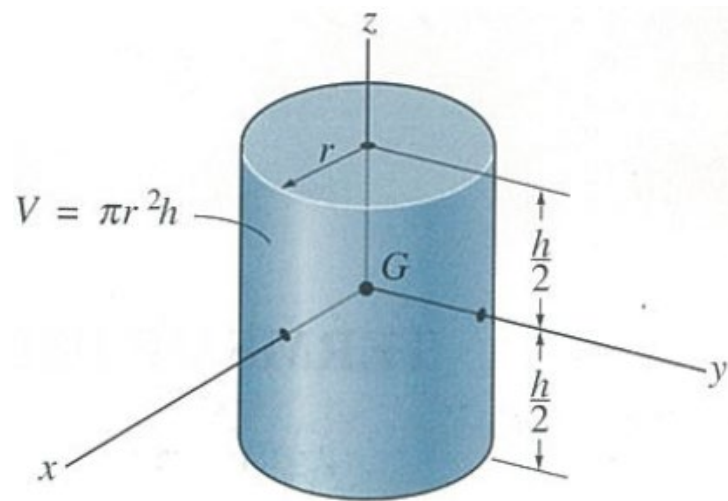
$$I_{GG} + md^2 = I_{ZZ} = mk^2$$

$$\therefore k > d$$



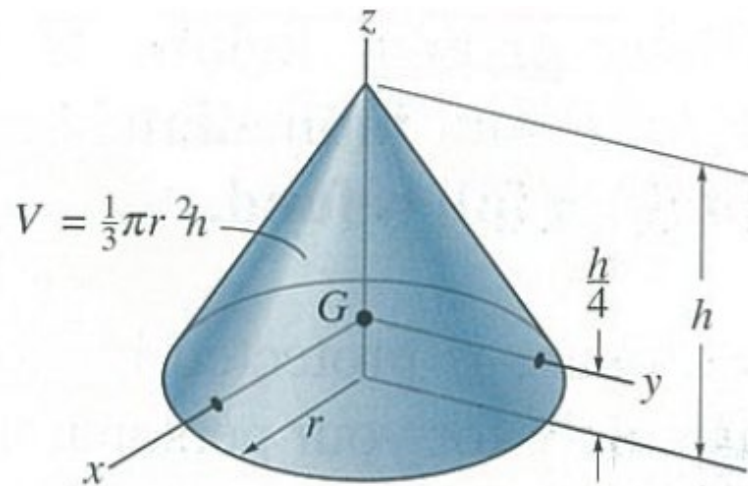
– Moment of inertia of typical parts (*mass uniformly distributed / homogenous*)





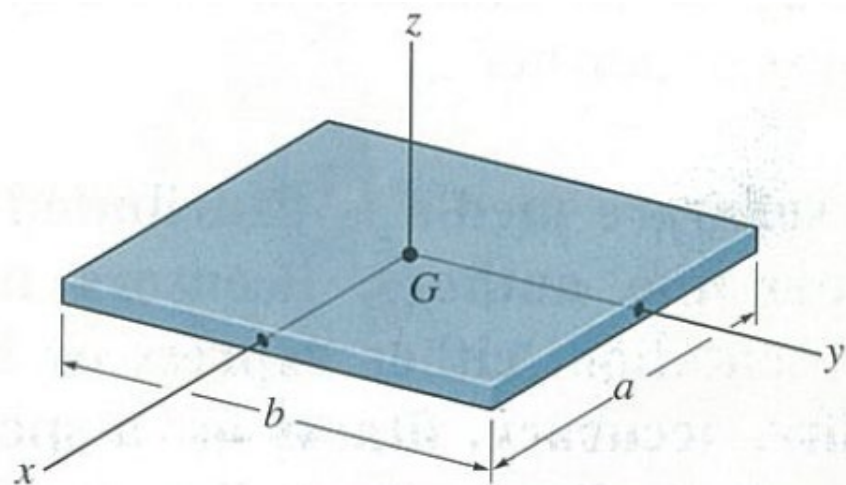
Cylinder

$$I_{xx} = I_{yy} = \frac{1}{12} m(3r^2 + h^2) \quad I_{zz} = \frac{1}{2} mr^2$$



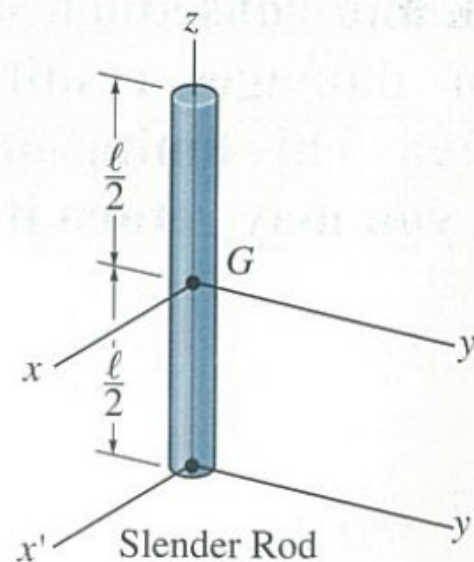
Cone

$$I_{xx} = I_{yy} = \frac{3}{80} m(4r^2 + h^2) \quad I_{zz} = \frac{3}{10} mr^2$$



Thin plate

$$I_{xx} = \frac{1}{12} mb^2 \quad I_{yy} = \frac{1}{12} ma^2 \quad I_{zz} = \frac{1}{12} m(a^2 + b^2)$$

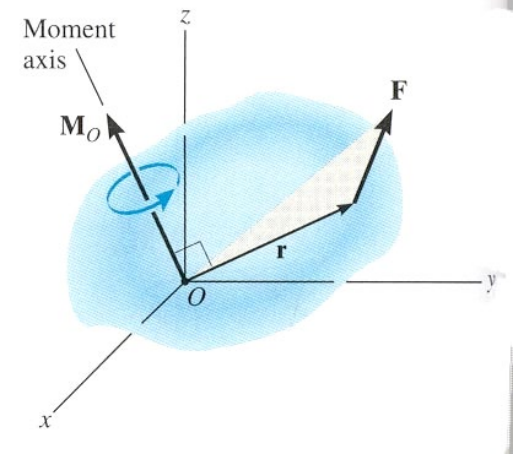


Slender Rod

$$I_{xx} = I_{yy} = \frac{1}{12} m\ell^2 \quad I_{x'x'} = I_{y'y'} = \frac{1}{3} m\ell^2 \quad I_{z'z'} = 0$$

• Moment of force (torque)

Using the vector cross product, $\mathbf{M}_O = \mathbf{r} \times \mathbf{F}$.



$$\mathbf{F}_1 = \{100 \mathbf{i} - 120 \mathbf{j} + 75 \mathbf{k}\} \text{ kN}$$

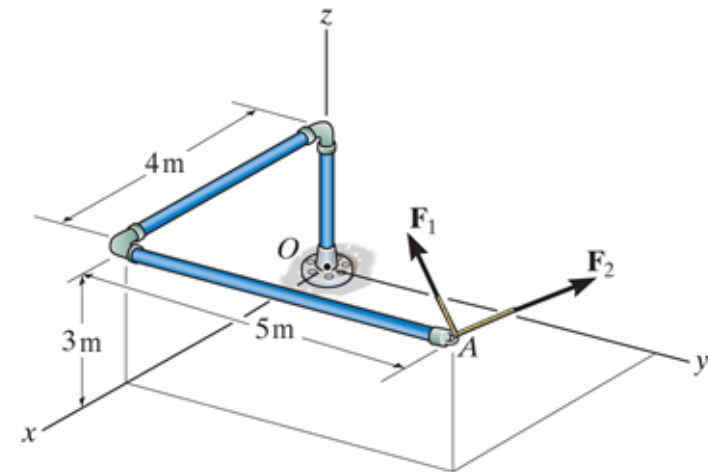
$$\mathbf{F}_2 = \{-200 \mathbf{i} + 250 \mathbf{j} + 100 \mathbf{k}\} \text{ kN}$$

$$\mathbf{F} = \mathbf{F}_1 + \mathbf{F}_2 = \{-100 \mathbf{i} + 130 \mathbf{j} + 175 \mathbf{k}\} \text{ kN}$$

$$\mathbf{r}_{OA} = \{4 \mathbf{i} + 5 \mathbf{j} + 3 \mathbf{k}\} \text{ m}$$

$$\mathbf{M}_O = \mathbf{r}_{OA} \times \mathbf{F}$$

$$= \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 4 & 5 & 3 \\ -100 & 130 & 175 \end{vmatrix} = \{485 \mathbf{i} - 1000 \mathbf{j} + 1020 \mathbf{k}\} \text{ kN m}$$

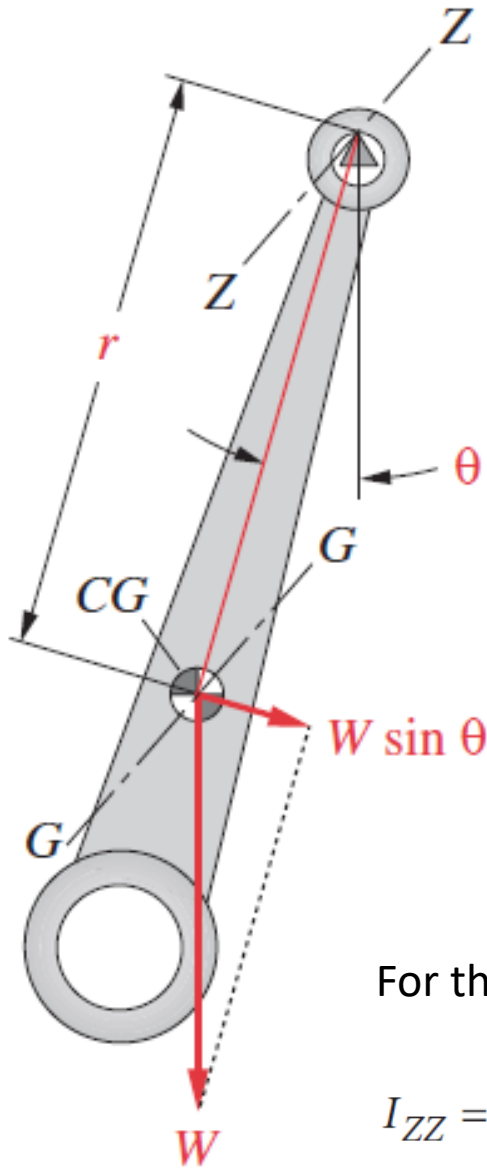


- **Newton's 2nd Law of rotational motion**

$$\mathbf{T} = I \alpha$$

$\mathbf{T}$ is the resultant torque *about the mass centre*, α is angular acceleration, and I is the mass moment of inertia *about an axis through the COM*.

• Measurement of MOI



$$\tau_{ZZ} = I_{ZZ} \alpha$$



$$-(W \sin \theta) r = I_{ZZ} \frac{d^2 \theta}{dt^2}$$

For small values of θ , $\sin \theta = \theta$



$$\frac{d^2 \theta}{dt^2} = -\frac{W r}{I_{ZZ}} \theta$$



$$\theta = \theta_{\max} \left(\cos \sqrt{\frac{W r}{I_{ZZ}}} t \right)$$

For the oscillation period $\tau \Rightarrow \sqrt{\frac{W r}{I_{ZZ}}} \tau = 2\pi \Rightarrow I_{ZZ} = W r \left(\frac{\tau}{2\pi} \right)^2$

$$I_{ZZ} = I_{GG} + m r^2 \Rightarrow I_{GG} = W r \left(\frac{\tau}{2\pi} \right)^2 - \frac{W}{g} r^2$$

• Work, Energy and Power

Work: *the dot product of force and displacement,* $W = \mathbf{F} \cdot \mathbf{R}$

Power: the time rate of change of energy, $P = \frac{dE}{dt}$

Kinetic energy: $\frac{1}{2}mv^2$ (translational motion)

$\frac{1}{2}I\omega^2$ (rotational motion around an axis)

Potential energy: mgh (gravitation)

$\frac{1}{2}kx^2$ (spring)

• Principle of D'Alembert

Dynamic equilibrium equations

$$\sum \underline{\mathbf{F}} - \underline{m\mathbf{a}} = 0$$

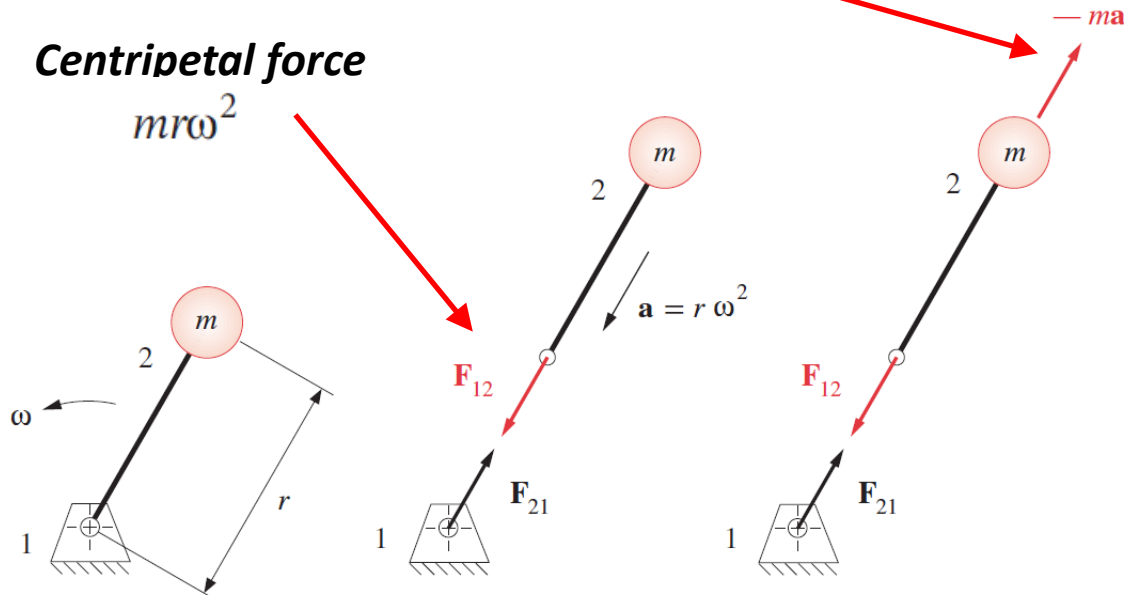
Initial Force

$$\sum \underline{\mathbf{T}} - \underline{I\alpha} = 0$$

Initial Torque

Example: Centripetal and centrifugal forces

Centrifugal force (initial force according to the principle of d'Alembert)



• Energy methods – *Virtual Work*

The rate of change of energy in the system at any instant must balance between that which is **externally supplied** and that which is **stored within the system** (neglecting losses).

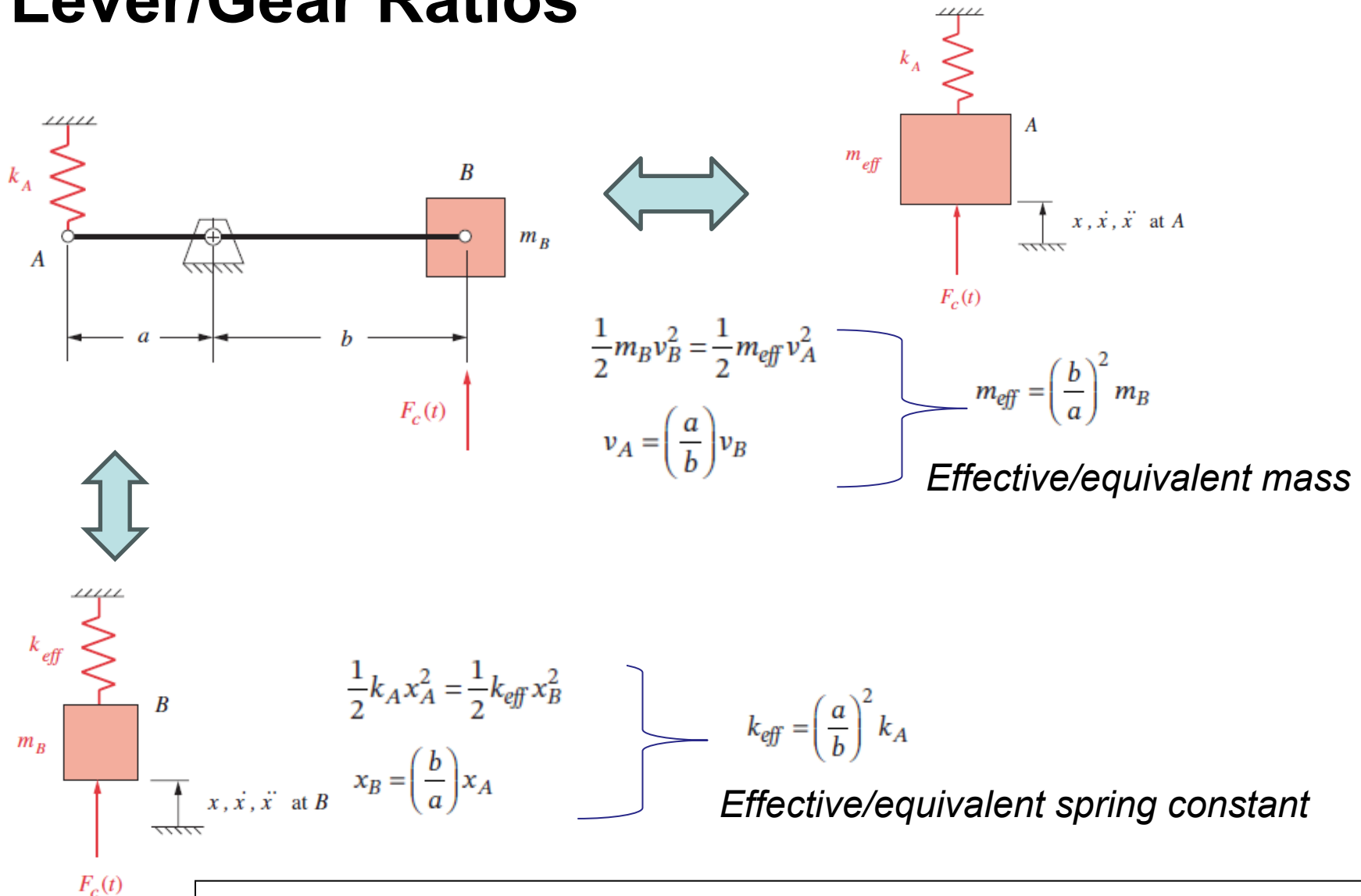
For a linkage of n links (Link1 is the ground),

$$\underbrace{\sum_{k=2}^n \mathbf{F}_k \cdot \mathbf{v}_k + \sum_{k=2}^n \mathbf{T}_k \cdot \omega_k}_{\text{External supplied}} = \underbrace{\sum_{k=2}^n m_k \mathbf{a}_k \cdot \mathbf{v}_k + \sum_{k=2}^n I_k \alpha_k \cdot \omega_k}_{\text{Stored within the system}}$$

Virtual work equation:

$$\sum_{k=2}^n \mathbf{F}_k \cdot \mathbf{v}_k + \sum_{k=2}^n \mathbf{T}_k \cdot \omega_k - \sum_{k=2}^n m_k \mathbf{a}_k \cdot \mathbf{v}_k - \sum_{k=2}^n I_k \alpha_k \cdot \omega_k = 0$$

- **Equivalent (*Energy Wise*) Systems via Lever/Gear Ratios**



The same principle applicable to the effective moment of inertia

- Linear equation in matrix form

$$\begin{cases} 2x_1 + 3x_2 - 2x_3 = 7 \\ -x_2 + x_1 - 3x_3 = 5 \end{cases}$$

$$\begin{cases} 2x_1 + 3x_2 - 2x_3 = 7 \\ x_1 - x_2 - 3x_3 = 5 \end{cases}$$



$$\begin{pmatrix} 2 & 3 & -2 \\ 1 & -1 & -3 \end{pmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 7 \\ 5 \end{bmatrix}$$

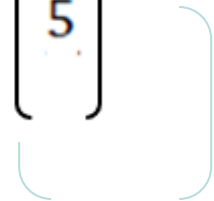
$$\begin{cases} 3x_2 - 2x_3 = 7 \\ x_1 - 3x_3 = 5 \end{cases}$$



$$\begin{cases} 0x_1 + 3x_2 - 2x_3 = 7 \\ x_1 + 0x_2 - 3x_3 = 5 \end{cases}$$



$$\begin{pmatrix} 0 & 3 & -2 \\ 1 & 0 & -3 \end{pmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 7 \\ 5 \end{bmatrix}$$



IPS/SI Units

(1.10: Pages 17 to 20)

IPS: U.S. inch-pound-second system

SI: Systems International

FPS: U.S. foot-pound-second system

U.S. Customary Units:
IPS and FPS

Variables and Units

Base Units in Boldface – Abbreviations in ()

Variable	Symbol	ips unit	SI unit
Force	F	pounds (lb)	newtons (N)
Length	l	Inches (in)	meters (m)
Time	t	seconds (sec)	seconds (sec)
Mass	m	lb–sec ² /in = bl	kilograms (kg)
<u>Weight</u>	W	pounds (lb)	newtons (N)
Velocity	v	in/sec	m/sec
Acceleration	a	in/sec ²	m/sec ²
Jerk	j	in/sec ³	m/sec ³
Angle	θ	degrees (deg)	degrees (deg)
Angle	θ	radians (rad)	radians (rad)
Angular velocity	ω	rad/sec	rad/sec
Angular acceleration	α	rad/sec ²	rad/sec ²
Angular jerk	φ	rad/sec ³	rad/sec ³
<u>Torque</u>	T	lb–in	N–m
<u>Mass moment of inertia</u>	I	lb–in–sec ²	<u>N–m–sec²</u> (kg - m ²)
Energy	E	in–lb	joules (J)
Power	P	in–lb/sec	watts (W)
Volume	V	in ³	m ³
Weight density	γ	lb/in ³	N/m ³
Mass density	ρ	bl/in ³	kg/m ³

Units Conversion

From U.S. Customary Units to Metric Units

1 Blob (bl)	=	175.127	Kilograms (kg)
1 Cubic inch (in ³)	=	16.387	Cubic centimeters (cc)
1 Foot (ft)	=	0.304 8	Meter (m)
1 Horsepower (hp)	=	745.699	Watts (W)
1 Inch (in)	=	0.025 4	Meter (m)
1 Mile, U.S. statute (mi)	=	1 609.344	Meters (m)
1 Pound force (lb)	=	4.448 2	Newtons (N)
	=	444 822.2	Dynes
1 Pound mass (lbm)	=	0.453 6	Kilogram (kg)
1 Pound-foot (lb-ft)	=	1.355 8	Newton-meter (N-m)
	=	1.355 8	Joules (J)
1 Pound-foot/second (lb-ft/sec)	=	1.355 8	Watts (W)
1 Pound-inch (lb-in)	=	0.112 8	Newton-meter (N-m)
	=	0.112 8	Joule (J)
1 Pound-inch/second (lb-in/sec)	=	0.112 8	Watt (W)
1 Pound/foot ² (lb/ft ²)	=	47.880 3	Pascals (Pa)
1 Pound/inch ² (lb/in ²), (psi)	=	6 894.757	Pascals (Pa)
1 Revolution/minute (rpm)	=	0.104 7	Radian/second (rad/s)
1 Slug (sl)	=	14.593 9	Kilograms (kg)
1 Ton, short (2000 lbm)	=	907.184 7	Kilograms (kg)

Units Conversion

Between U.S. Customary Units

1 Blob (bl)	=	12	Slugs (sl)
1 Blob (bl)	=	386	Pounds mass (lbm)
1 Foot (ft)	=	12	Inches (in)
1 Horsepower (hp)	=	550	Pound-feet/second (lb-ft/sec)
1 Knot	=	1.151 5	Miles/hour (mph)
1 Mile, U.S. statute (mi)	=	5 280	Feet (ft)
1 Mile/hour	=	1.4667	Feet/second (ft/sec)
1 Pound force (lb)	=	16	Ounces (oz)
1 Pound mass (lbm)	=	0.0311	Slug (sl)
1 Pound-foot (lb-ft)	=	12	Pound-inches (lb-in)
1 Pound-foot/second (lb-ft/sec)	=	0.001 818	Horsepower (hp)
1 Pound-inch (lb-in)	=	0.083 3	Pound-foot (lb-ft)
1 Pound-inch/second (lb-in/sec)	=	0.021 8	Horsepower (hp)
1 Pound/inch ² (lb/in ²), (psi)	=	144	Pounds/foot ² (lb/ft ²)
1 Radian/second (rad/sec)	=	9.549	Revolutions/minute (rpm)
1 Slug (sl)	=	32.174	Pounds mass (lbm)
1 Ton, short	=	2000	Pounds mass (lbm)